Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems

Project Report Submitted
in partial fulfillment of the requirements for the award of the

Bachelor of Technology

in

Computer Science and Engineering

by

Subhranshu Panda
(B122117)

Shreyansh Gupta
(B122109)

Under the supervision of
Prof. Bharati Mishra

Department of Computer Science and Engineering
International Institute of Information Technology, Bhubaneswar
May 2026

Approval of the Viva-Voce Board

Date: ______________, 2026

Certified that the report entitled "Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems" submitted by Subhranshu Panda (B122117) and Shreyansh Gupta (B122109) to International Institute of Information Technology Bhubaneswar in partial fulfillment of the requirements for the award of the Bachelor of Technology in Computer Science Engineering under the BTech Programme has been accepted by the examiners during the viva-voce examination held today.

______________________	______________________
(Supervisor)	(Panel Head)
______________________	______________________
(Internal Examiner 1)	(Internal Examiner 2)

____

(External Examiner)

Certificate

This is to certify that the report entitled "Building the Glass Box : A Human-Centered Framework for Explainable AI in Cyber-Physical Systems" submitted by Subhranshu Panda (B122117) and Shreyansh Gupta (B122109) to International Institute of Information Technology Bhubaneswar is a record of bonafide project work under my supervision, and the report is submitted for the award of the Bachelor of Technology in Computer Science and Engineering.

Prof. Bharati Mishra
Supervisor
Department of Computer Science and Engineering
IIIT Bhubaneswar

Declaration

We, Subhranshu Panda (B122117) and Shreyansh Gupta (B122109), hereby declare that this project report entitled "Building the Glass Box : A Human-Centered Framework for Explainable AI in Cyber-Physical Systems" submitted to International Institute of Information Technology Bhubaneswar is a record of an original work done by us under the guidance of Prof. Bharati Mishra, and this project work has not formed the basis for the award of any Degree or diploma/associate ship/fellowship and similar project if any.

Subhranshu Panda	**Shreyansh Gupta**
(B122117)	(B122109)

Acknowledgment

We would like to express our sincere gratitude to our project guide, Prof. Bharati Mishra, for their invaluable guidance, support, and encouragement throughout this research. Their expertise and insightful feedback were instrumental in shaping this report.

We also extend our thanks to the panel members for their time and constructive criticism, which helped us refine our work.

Our gratitude also goes to IIIT Bhubaneswar for providing us with the necessary resources and a conducive environment for our research.

Finally, we would like to thank our colleagues, friends, and families for their unwavering support and patience during this entire process.

Subhranshu Panda
(B122117)

Shreyansh Gupta
(B122109)

Abstract

The integration of Artificial Intelligence (AI) and Cyber-Physical Systems (CPS) is driving a new industrial transformation. However, the "black box" nature of high-performance AI models creates catastrophic risks in safety-critical systems, leading to a crisis in trust and accountability. Explainable AI (XAI) emerges as the essential solution to provide transparency and human-interpretable explanations for AI-driven decisions. Building upon our theoretical framework, this report presents the complete empirical implementation of a human-centered, context-aware XAI-CPS. We developed a "Glass Box" software prototype simulating anomalies across two safety-critical domains: a Smart Water Treatment System and a Smart Power Grid. Utilizing a secure, offline Large Language Model (Llama 3.2) orchestrated via a multi-agent framework (Microsoft AutoGen), the system successfully contrasts traditional context-agnostic explanations with our proposed context-aware XAI. Crucially, to overcome the bottleneck of manual human testing, we pioneer an Automated Human-Centered Evaluation using an LLM-as-a-Judge architecture. By simulating expert domain personas, the system objectively proves the framework's capability to significantly enhance Trust, Actionability, and Reasonableness in industrial crisis management.

Contents

1 Introduction

1.1 The Rise of AI in Cyber-Physical Systems

Cyber-Physical Systems (CPS) represent the profound integration of computation, networking, and physical processes. They form the backbone of critical modern infrastructure, including smart manufacturing, autonomous transportation, and healthcare systems [2, 3]. The advent of Industry 4.0 and the ongoing transition towards Industry 5.0 heavily rely on deploying advanced Deep Learning models within these networks to achieve predictive maintenance, real-time optimization, and autonomous decision-making [26].

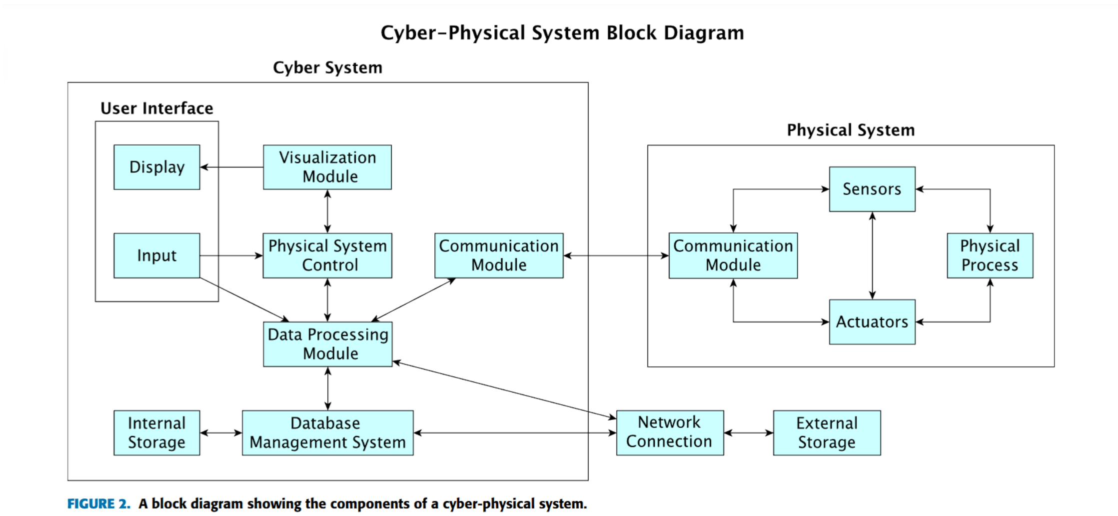

Figure 1: A block diagram showing the components of a cyber-physical system [1].

1.2 The "Black Box" Crisis

Despite their high predictive accuracy, state-of-the-art deep learning models inherently function as opaque "black boxes." While this opacity may be acceptable in low-risk applications, it introduces catastrophic risks in safety-critical CPS environments [25]. When an autonomous system makes a critical error or an industrial plant experiences an unpredicted anomaly, human operators cannot answer the fundamental question: *"Why did the AI make this decision?"* This lack of algorithmic transparency creates a severe deficit in trust, safety validation, and operational accountability.

1.3 The Need for Explainable AI (XAI)

Explainable AI (XAI) has emerged as the essential solution to this crisis. XAI provides the necessary "glass box" methodologies to produce human-interpretable explanations for complex AI behaviors, transforming opaque algorithms into accountable, auditable tools [24]. However, successfully implementing XAI in CPS requires moving beyond mere algorithmic transparency to true environmental context-awareness, which forms the core focus of this project's empirical implementation.

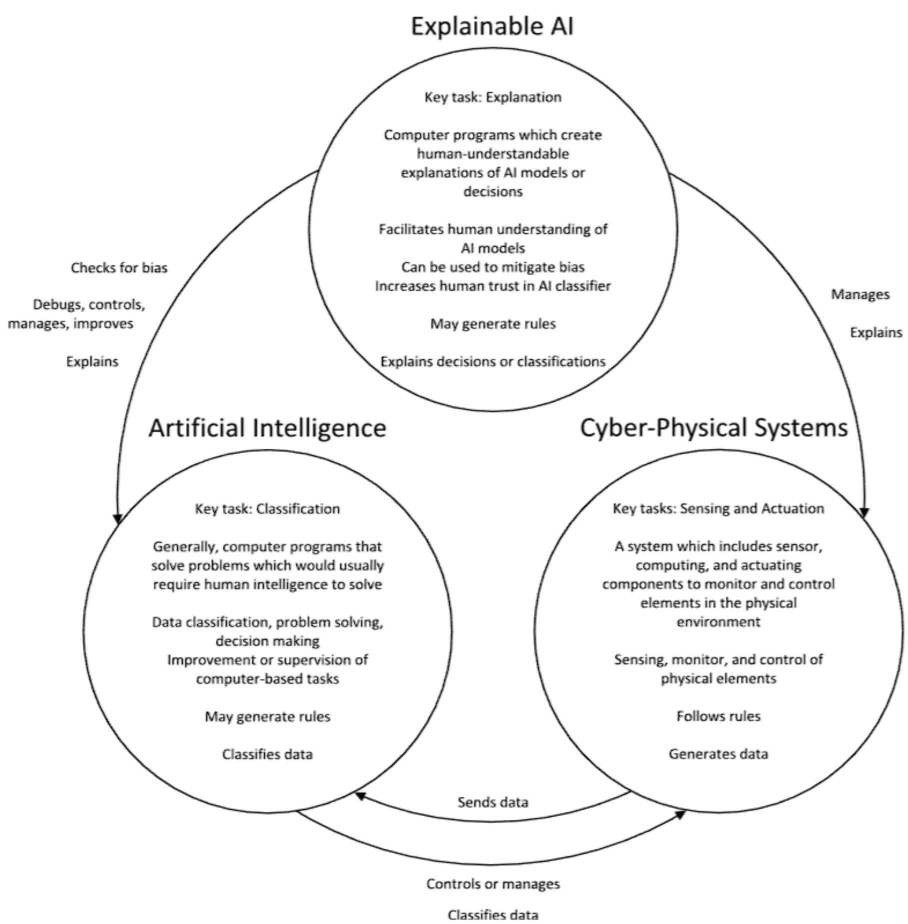

Figure 2: A comparison diagram showing the characteristics and connections between AI for classification, XAI, and cyber-physical systems [1].

1.4 Scope of the Final Project

While our mid-semester research established the theoretical necessity of context-aware XAI, this final project report transitions that theory into a fully functional, empirically validated software architecture. We expand the scope of the prototype to encompass dual industrial domains—a Smart Water Treatment Facility and a Smart Power Grid—proving the framework's versatility. Furthermore, we address the critical bottleneck of evaluating XAI by engineering an automated "LLM-as-a-Judge" pipeline. By deploying autonomous Expert Evaluator agents, this project successfully quantifies the operational superiority of context-aware explanations, delivering a comprehensive, enterprise-ready "Glass Box" dashboard for modern Industry 5.0 infrastructure.

2 Literature Survey

The integration of Explainable AI (XAI) within Cyber-Physical Systems (CPS) has garnered significant academic attention, primarily focusing on resolving the "black box" nature of deep learning models in safety-critical domains such as smart manufacturing, healthcare, and autonomous grids. However, a critical review of the current state-of-the-art reveals a persistent over-reliance on traditional, context-agnostic XAI techniques (e.g., SHAP, LIME). These methods evaluate internal model features in a vacuum, completely ignoring the dynamic external physical environments in which these machines operate.

To identify the specific technical and methodological gaps addressed by our empirical prototype, Table 1 summarizes the key contributions and critical limitations of prominent recent literature in this domain.

Reference/Author	**CPS Domain**	**XAI Methodology**	**Critical Limitation / Gap**
Wickramasinghe et al. [1]	Industrial IoT	Unsupervised ML feature extraction	Lacks human-centric evaluation metrics; purely algorithmic focus.
Oliveira et al. [4]	Chemical Industry	Fault diagnosis and optimization	Context-agnostic; does not correlate mechanical faults with external environments.
Alimonda et al. [6]	Medical CPS	CLIX-M evaluation framework	Evaluated only in theoretical simulations; lacks empirical real-world testing.
Almuqren et al. [8]	CPS Security (IDS)	Feature weighting via SHAP/LIME	Explanations are isolated to network traffic, ignoring physical sensor states.
Our Proposed Work	Smart Water Treatment	**Multi-Agent Context-Aware LLMs**	**Bridges the gap by correlating internal telemetry with external environmental variables.**

Table 1: Summary of State-of-the-Art XAI in CPS and Identified Research Gaps

This comprehensive analysis confirms that while algorithmic transparency is improving, the industry desperately needs a *context-aware* framework evaluated through empirical, human-centered metrics—which directly motivates our 8th-semester prototype implementation. The intersection of Artificial Intelligence (AI) and Cyber-Physical Systems (CPS) has been extensively studied, primarily focusing on optimizing algorithmic performance for predictive maintenance and anomaly detection [2, 10]. However, the operational deployment of these systems in safety-critical domains introduces complex human-machine interaction challenges that pure performance metrics cannot capture [4, 25].

2.1 The "Black Box" Crisis in CPS

Modern Deep Learning models excel at identifying latent patterns within massive CPS telemetry datasets [1, 27]. However, their internal architectures render them opaque to human operators. In industrial settings, this "black box" paradigm is a critical vulnerability [11]. When an AI system flags a catastrophic anomaly without providing an intelligible rationale, operators are forced to either blindly trust the machine or ignore the warning entirely. Research indicates that this lack of transparency directly degrades operational safety, violates emerging regulatory compliance standards, and severely bottlenecks the transition to human-centric Industry 5.0 paradigms [12, 26].

2.2 Limitations of Traditional XAI (LIME and SHAP)

To address algorithmic opacity, Explainable AI (XAI) frameworks such as Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have become industry standards [15, 24]. These feature-attribution methods mathematically approximate the behavior of a black-box model, outputting weights that indicate which input features drove a specific prediction.

While mathematically rigorous, these traditional methods suffer from a severe operational limitation in CPS environments: the *context-awareness gap* [5, 6]. Standard XAI evaluates telemetry in a vacuum. For instance, if a water pressure sensor reports a sudden drop, traditional XAI will accurately flag the pressure variable as the root cause of the anomaly. However, it ignores external physical and virtual environments—such as severe localized weather patterns or network latency—that frequently trigger these internal sensor fluctuations. Consequently, traditional XAI often misdiagnoses context-driven anomalies as imminent mechanical failures, generating false positives that compromise actionability [7,13].

2.3 The Shift Toward Human-Centered Evaluation

The literature is increasingly calling for a paradigm shift from algorithm-centric XAI to Human-Centered Explainable AI (HCXAI) [1,11]. Recent studies argue that an explanation's quality should not be judged solely by mathematical fidelity to the base model, but by its utility to the end-user [16]. In the context of CPS, an explanation must be evaluated on its ability to build human **Trust**, demonstrate physical **Reasonableness**, and drive operational **Actionability** [26].

2.4 The Evaluation Bottleneck and Cross-Domain Generalizability

Despite the recognized need for HCXAI, the field currently suffers from two major systemic bottlenecks. First, *Domain Siloing*: most proposed XAI frameworks are over-fitted to a single specific use-case (e.g., only water treatment or only medical data), failing to prove cross-domain generalizability. Second, the *Evaluation Bottleneck*: executing formalized human-centered evaluation requires recruiting highly specialized domain experts to manually review hundreds of AI explanations, making large-scale iterative testing nearly impossible. This paper directly addresses both voids by validating our framework across dual industrial domains (Water and Power) and introducing a multi-agent architecture capable of executing automated expert evaluations at scale.

3 Motivation

The primary motivation for this research stems from the urgent need to operationalize Explainable AI (XAI) within safety-critical Cyber-Physical Systems (CPS) as they evolve toward the human-machine collaboration standards of Industry 5.0 [26]. While theoretical models of XAI have advanced significantly, their practical deployment in industrial settings remains severely stalled.

Traditional feature-attribution methods operate in a context-agnostic vacuum, rendering their explanations mathematically accurate but operationally inadequate [15, 24]. In high-stakes domains—such as smart water treatment facilities or power grids—a false positive generated by a lack of environmental context results in costly, unnecessary maintenance downtime and a rapid erosion of operator trust [5]. Therefore, there is a critical need to bridge the gap between algorithmic theory and operational reality by developing XAI systems that actually understand the physical and virtual environments in which they operate. Furthermore, this transition must be achieved without exposing proprietary industrial telemetry to the severe cybersecurity vulnerabilities inherent in cloud-based AI APIs [9].

4 Research Objectives

To address these critical operational gaps, the primary goal of this research is to design, implement, and empirically validate a Context-Aware, Human-Centered XAI Framework. Evolving from our initial theoretical proposals, this final implementation achieves the following core objectives:

- **Develop a Context-Aware Reasoning Engine:** To formulate a framework that actively correlates internal mechanical telemetry (e.g., pressure, vibration, voltage) with dynamic external environmental variables (e.g., weather patterns, network latency), thereby eliminating context-agnostic false positives.
- **Ensure Cybersecurity-by-Design:** To architect a 100% offline, multi-agent explanation pipeline utilizing localized Large Language Models (Llama 3.2 via Ollama) that guarantees strict data privacy for sensitive industrial infrastructure.
- **Prove Cross-Domain Scalability:** To transition beyond single-domain testing by deploying our framework across two distinct, synthesized 1,000-sample telemetry streams: a Smart Water Treatment Facility and a Smart Power Grid.
- **Automate Human-Centered Evaluation:** To pioneer an LLM-as-a-Judge methodology using Microsoft AutoGen. By simulating highly specialized domain expert personas (e.g., "Senior Grid Operations Engineer"), the system objectively scores AI explanations based on the metrics most crucial to human operators: Trust, Reasonableness, and Actionability.

5 System Architecture and Methodology

The widespread, trusted adoption of XAI in CPS is currently blocked by two fundamental gaps: a **Technical Gap** (lack of context-awareness) [1,10] and an **Evaluation Gap** (lack of formalized human-centered standards) [1,14,15]. To solve this, we implemented a three-phase methodological framework that integrates these solutions directly into a fully functional software prototype.

5.1 Phase 1: System Design and Requirements Analysis

This foundational phase frames explainability not as an add-on, but as a core system requirement from inception [1].

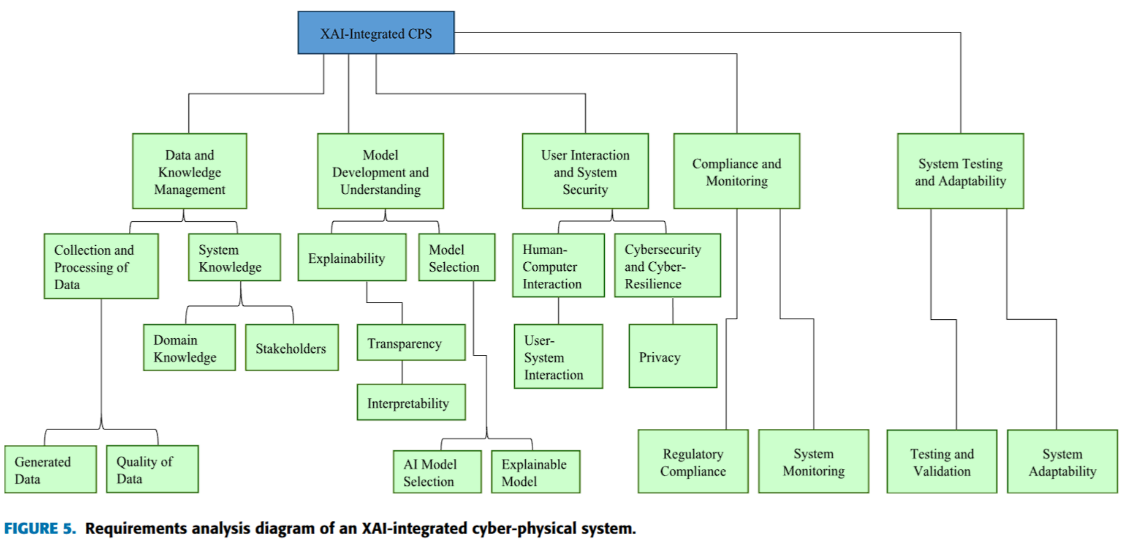

Figure 3: Requirements analysis diagram of an XAI-integrated cyber-physical system [1].

The design process mandates four critical pillars:

- **Data and Knowledge Management:** Ensuring representative datasets and establishing formal methods to incorporate human domain physics into the AI model.
- **Model Selection:** Balancing predictive performance with interpretability, consciously pairing necessary "black box" models with domain-specific XAI frameworks [1,7].
- **Human-Computer Interaction (HCI):** Defining the target user (e.g., engineer vs.

manager) and tailoring the explanation's interface and modality to their cognitive load [1].

- **Cybersecurity-by-Design:** Integrating principles from frameworks like the Explainable Resiliency Graph (ERG) to model how the system will explain cyber-physical attacks [1, 8].

5.2 Phase 2: Automated Human-Centered Evaluation (LLM-as-a-Judge)

Addressing the "Evaluation Gap" requires judging an explanation's quality not purely by algorithmic fidelity, but by its operational utility [1, 15]. Standard AI metrics like Accuracy or F1-scores measure if a model is correct, but in a Cyber-Physical System, correctness alone is insufficient. Similarly, traditional XAI metrics (like SHAP or LIME) output feature weight numbers, not human-readable reasoning. The entire goal of XAI in CPS is to assist human decision-making under time pressure; therefore, we must evaluate the explanation's usefulness to a human operator.

To bypass the bottleneck of recruiting scarce domain experts for manual testing, we pioneered an **Automated Human-Centered Evaluation (HCE)** pipeline utilizing an LLM-as-a-Judge architecture. We synthesize the following three core metrics as our standard for evaluation:

- **Trust ("Can I rely on this AI?"):** In safety-critical systems, an operator will ignore an AI they do not trust. If the AI suggests a "mechanical failure" during a visible lightning storm, trust is immediately lost. This metric measures whether the explanation feels credible and consistent with reality, serving as the foundation for human-AI collaboration.

- **Reasonableness ("Does this explanation make logical sense?"):** An explanation can be technically correct but logically unreasonable to a domain expert. This checks if the logic of the explanation holds up to expert scrutiny and aligns with established domain knowledge and engineering physics.

- **Actionability ("What do I actually do now?"):** An explanation is useless if it does not guide a concrete next step. Telling an operator "sensors are abnormal" is inadequate; diagnosing that a "storm caused a voltage sag—activate the backup transformer" drives immediate resolution. Actionability measures whether the output successfully guides a physical response.

5.3 Phase 3: Dual-Domain Empirical Testing and Prototype Implementation

To transition our framework from theoretical design to empirical validation, we developed a functional "Glass Box" software prototype. To prove cross-domain generalizability, we expanded the scope of our prototype to evaluate anomalies across two distinct safety-critical CPS environments.

5.3.1 The Data Challenge: Justification for Data Synthesis

A primary challenge in validating XAI for Cyber-Physical Systems is the acquisition of functional datasets. While real-world telemetry is traditionally preferred, accessing raw, labeled data from modern CPS is practically impossible due to two critical constraints:

1. **National Security and Air-Gapping:** Smart Power Grids and Water Treatment Plants constitute critical national infrastructure. Their Supervisory Control and Data Acquisition (SCADA) networks are strictly proprietary, classified, and air-gapped to prevent cyber-terrorism, making real telemetry unavailable to public researchers.

2. **The Contextual Labeling Deficit:** Even when real operational datasets are available, they suffer from a severe labeling deficit. They typically only record internal mechanical states (the "what") and rarely possess perfectly synchronized labels for external environmental events (weather, network latency). Testing a *Context-Aware* XAI framework requires a dataset where external triggers are perfectly mapped to internal deviations.

To overcome this, we engineered a physics-grounded Python synthesis script (utilizing NumPy and Pandas) to generate two 1,000-sample time-series datasets representing 15-minute operational intervals. The telemetry is bound by Gaussian distributions centered on real-world engineering standards, deliberately isolating mechanical failures from contextual anomalies to rigorously stress-test the XAI's causal reasoning.

5.3.2 Scenario A: Smart Water Treatment System

The first synthesized dataset represents an IoT-enabled Smart Water Treatment Facility.

- **Base Telemetry:** Normal operations idle at a pressure of ≈ 50 psi and pump vibration of ≈ 2.0 mm/s. Normal network latency is bounded at ≈ 20 ms.
- **Mechanical Failures:** Isolated equipment breakdowns where pressure drops to 25 psi and vibration spikes to 8.0 mm/s, with no external weather triggers.
- **Contextual Anomalies:** A heavy storm system hits the facility's geographic area, causing severe network latency (spiking up to 200 ms). This data lag causes the pump to overwork, dropping pressure to 28 psi and raising vibration to 6.5 mm/s.

5.3.3 Scenario B: Smart Power Grid

To validate domain-agnostic scalability, a second 1,000-sample dataset was synthesized representing a high-voltage Smart Power Grid:

- **Base Telemetry:** Normal operations maintain a strict tolerance of ≈ 132 kV Voltage, ≈ 400 A Current, and a highly stable ≈ 50.0 Hz Frequency.
- **Mechanical Failures:** True physical faults, such as a transformer failure or line fault, causing voltage to collapse to ≈ 62 kV and current to spike to ≈ 900 A.
- **Contextual Anomalies:** A severe regional heatwave causing massive thermal stress and unpredictable HVAC electrical loads, resulting in voltage sags (≈ 108 kV) and frequency dips (≈ 49.2 Hz) driven entirely by external demand, not hardware degradation.

5.3.4 System Architecture: Detection and Multi-Agent Orchestration

To process this dual-domain telemetry securely, we engineered a local, multi-agent AI pipeline. The architecture is intentionally divided into two distinct steps to mirror real-world industrial SCADA systems:

Step 1: Rule-Based Anomaly Detection

Before any AI agents are triggered, the system flags anomalies using hard engineering thresholds (e.g., Voltage < 110 kV and Frequency < 49.5 Hz for the Power Grid; Pressure < 35 psi and Vibration > 4.5 mm/s for the Water Plant). While Machine Learning could be used for detection, rule-based thresholds are the established standard in industrial environments because engineers define safe operating ranges. Our primary contribution is not anomaly *detection*, but rather the *explanation* of why that anomaly occurred.

Step 2: Multi-Agent XAI Generation

Once an anomaly is flagged, the data is passed to the explanation engine. To address the cybersecurity vulnerabilities of cloud-based APIs, our prototype executes entirely locally, utilizing **Ollama** to run the **Llama 3.2** model orchestrated via **Microsoft AutoGen**.

The explanation pipeline is driven by three distinct autonomous agent personas:

1. **Agent 1a (Context-Agnostic Explainer):** Simulating traditional XAI, this agent receives only internal sensor values. It assumes mechanical failure by default without looking at the external context.

2. **Agent 1b (Context-Aware Explainer):** Representing our proposed framework, this agent receives sensor values *plus* external environmental context and network latency. It links sensor deviations to external triggers to generate a causally grounded explanation.

3. **Agent 2 (Expert Evaluator):** Utilizing the LLM-as-a-Judge framework, this agent adopts a specialized persona (e.g., "Senior Grid Operations Engineer") to objectively score both Agent 1a and Agent 1b explanations out of 5 based on Trust, Reasonableness, and Actionability.

Defining the Algorithm: Prompt Engineering as XAI

Traditional XAI relies on algorithms like SHAP or LIME to generate numerical feature weights. In our framework, the "algorithm" is advanced prompt engineering combined with LLM reasoning. By assigning strict personas, defining contextual data perimeters, and dictating output formats, we force the LLM to make its reasoning process explicit and transparent. The agent explains causality in plain language that an operator can verify and act upon. Moving beyond black-box scores to human-understandable reasoning is the true definition of Explainable AI.

5.3.5 The "Glass Box" Interface

A human-machine interface was constructed using **Streamlit** to serve as the central dashboard for the CPS facility manager. The dashboard features a Domain Selector, allowing operators to switch between the Smart Water Plant and the Power Grid. Figure 4 displays the real-time sensor telemetry where anomalies are actively tracked.

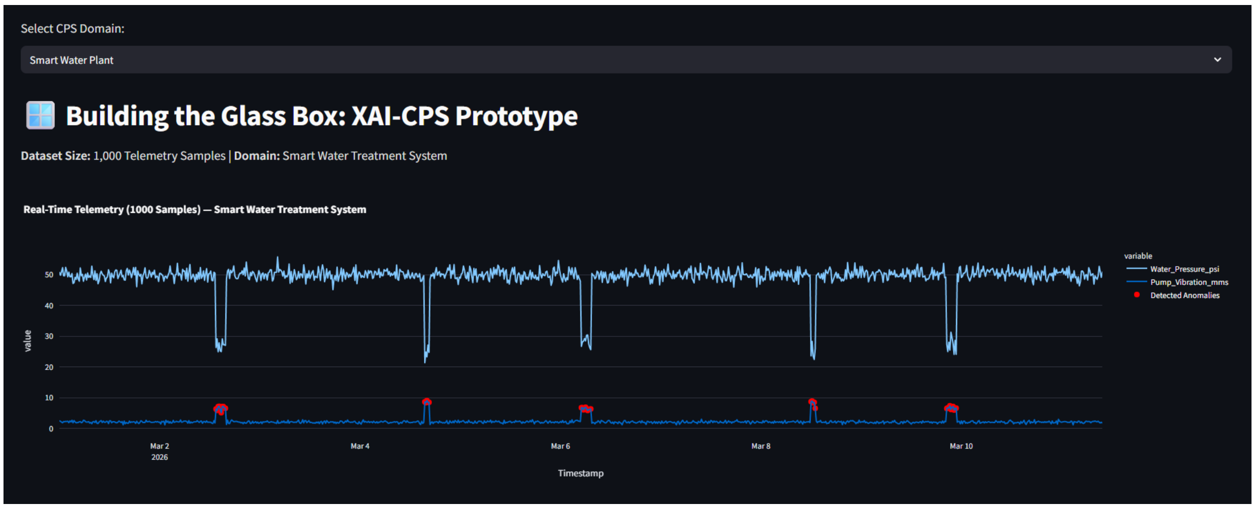

Figure 4: The "Glass Box" Streamlit dashboard displaying real-time CPS sensor telemetry and the dynamically detected anomalies across selected domains.

When the anomaly is triggered, the dashboard explicitly contrasts the two explanation paradigms side-by-side:

- **System A (Traditional XAI):** Provides a context-agnostic diagnosis, incorrectly suggesting a critical physical hardware failure based solely on internal telemetry (e.g diagnosing a broken transformer instead of recognizing heatwave load).
- **System B (Proposed Context-Aware XAI):** Successfully correlates the internal sensor deviations with the external weather/network context, correctly diagnosing the root environmental cause.

Immediately following the generation of the explanations, the **Expert Evaluator Agent** outputs its automated scorecards, providing a transparent, unbiased Quick Comparison Table (Figure 5) to assist the human operator in rapid crisis resolution.

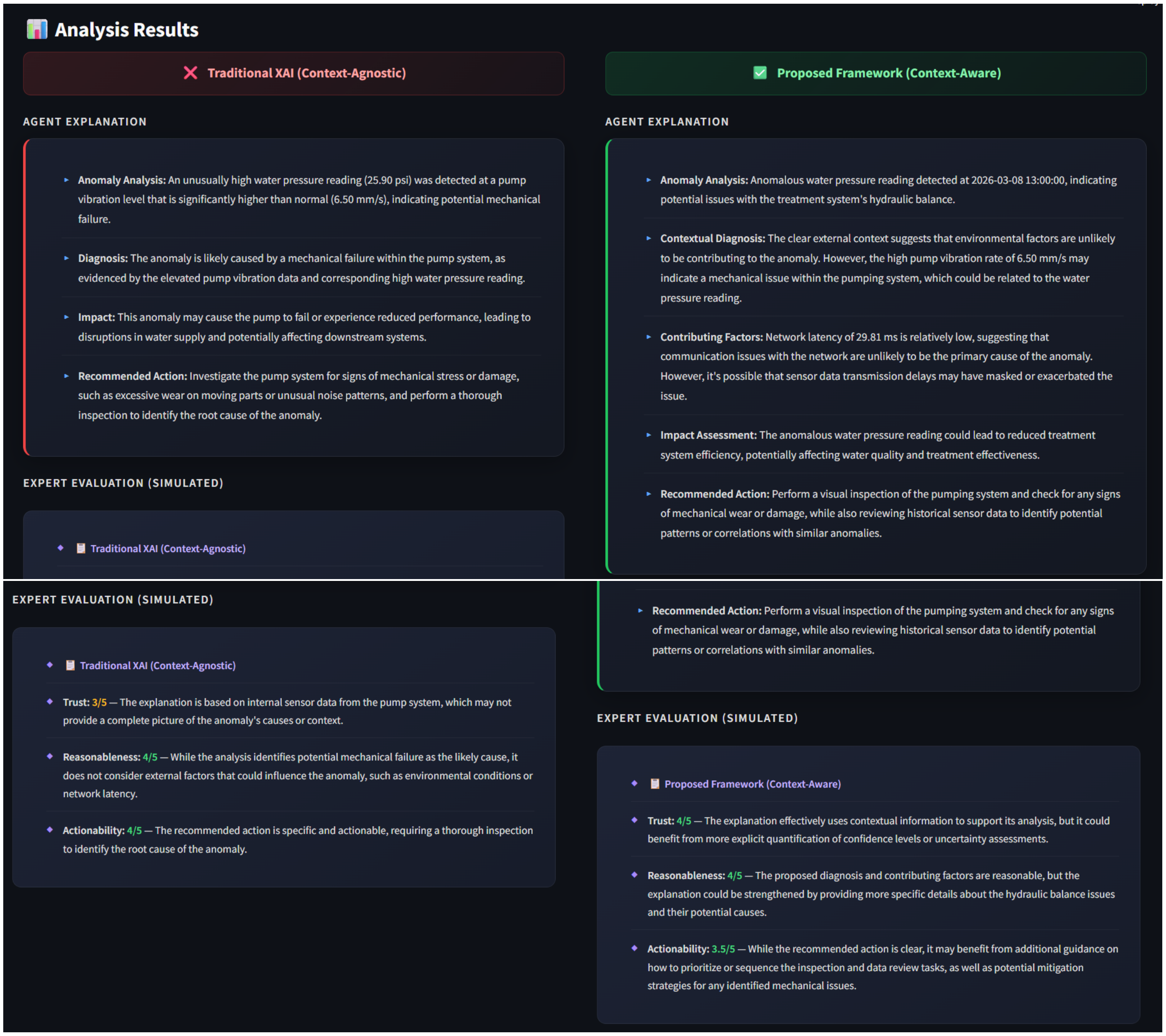

Figure 5: The multi-agent output interface. **Top:** Side-by-side empirical comparison contrasting traditional context-agnostic XAI with the proposed context-aware framework. **Bottom:** The automated LLM-as-a-Judge scorecards grading both explanations on Actionability, Reasonableness, and Trust.

6 Recommendations and Future Research Directions

While the empirical implementation of our dual-domain "Glass Box" prototype successfully demonstrates the power of context-aware XAI, there are still exciting opportunities to push this technology even further [25].

6.1 For System Development: Expanding the Multi-Agent Ecosystem

Our current prototype utilizes a three-agent architecture (CPS Monitor, XAI Explainer, and Expert Evaluator) orchestrated via Microsoft AutoGen. A natural next step is to expand this into a larger, decentralized ecosystem of highly specialized agents [1]. By integrating additional roles—such as a "Predictive Maintenance Agent" or an "Autonomous Safety Auditor"—running on local, offline LLMs, the CPS could achieve true self-explainability. In this future state, different AI nodes would autonomously cross-examine each other's reasoning and debate potential solutions before presenting a final, context-aware recommendation to the human operator.

6.2 For the Human-AI Interface: Multisensory Integration

Right now, our Streamlit dashboard relies entirely on visual graphs and text blocks. However, in a loud, high-stakes industrial environment, staring at a screen isn't always practical or safe. Future iterations of the prototype should leverage multimodal AI to deliver *multisensory* explanations [16]. For example, translating the AI's context-aware diagnosis into targeted haptic feedback (such as a vibrating smart-glove) or directional audio alerts. This would ensure the explanation seamlessly matches the operator's physical environment, heavily reducing cognitive load and actively preventing occupational burnout [26, 28].

6.3 For the Research Community: Validating Automated Evaluation

In this project, we pioneered an LLM-as-a-Judge architecture to overcome the slow, expensive process of having humans manually evaluate XAI. While this automated scoring of Trust, Reasonableness, and Actionability is incredibly efficient, the research community must take the next crucial step: rigorously validating these AI-generated scores against actual human responses [15]. Future studies should compare our automated LLM scorecards with objective human biometric data—such as eye-tracking or EEG scans of operators using the dashboard—to definitively prove that LLM evaluators accurately mirror actual human comprehension and trust in XAI-CPS [1].

7 Conclusion

7.1 Summary of Findings

The “black box” nature of complex AI models poses a catastrophic risk to safety-critical Cyber-Physical Systems, creating a profound barrier to trust and operational deployment [25]. Explainable AI (XAI) is the essential “glass box” solution to this crisis [11]. However, our foundational research identified that traditional XAI methods fail in real-world applications because they are predominantly context-agnostic—unable to explain the system’s interaction with its dynamic physical environment—and lack formalized, scalable evaluation standards [1, 15].

7.2 A Path Forward

To solve these critical gaps, this project successfully transitioned from a theoretical methodology to a fully functional, empirical implementation. We engineered a secure, 100% offline multi-agent XAI prototype using Llama 3.2 and Microsoft AutoGen, wrapped in a highly readable, human-centric Streamlit dashboard.

Crucially, this final project expanded beyond our initial scope to prove cross-domain scalability. By testing the framework against simulated anomalies in both a Smart Water Treatment Facility and a Smart Power Grid, the system successfully demonstrated that context-aware AI actively outperforms traditional methods by correlating internal mechanical and electrical deviations with external environmental variables. Furthermore, we successfully resolved the industry-wide human evaluation bottleneck by pioneering an LLM-as-a-Judge architecture. By deploying an autonomous Expert Evaluator agent, we proved that AI explanations can be objectively, rapidly, and accurately scored for Trust, Reasonableness, and Actionability.

Ultimately, by building and deploying this "Glass Box" system, we have proven that context-aware, trustworthy AI is not only practically achievable but also highly scalable across different industrial domains. This implementation provides a concrete pathway toward the vision of Industry 5.0—where intelligent technology is not just an opaque, unpredictable tool, but a transparent, secure, and truly collaborative partner for human operators [26].

References

[1] C. S. Wickramasinghe, K. Amarasinghe, D. L. Marino, C. Rieger, and M. Manic, "Explainable unsupervised machine learning for cyber-physical systems," *IEEE Access*, vol. 9, pp. 131824–131843, 2021.

[2] F. Hu, Y. Lu, A. V. Vasilakos, Q. Hao, R. Ma, Y. Patil, T. Zhang, J. Lu, X. Li, and N. N. Xiong, "Robust cyber-physical systems: Concept, models, and implementation," *Future Gener. Comput. Syst.*, vol. 56, pp. 449–475, Mar. 2016.

[3] J. Lee, B. Bagheri, and H.-A. Kao, "A cyber-physical systems architecture for Industry 4.0-based manufacturing systems," *Manuf. Lett.*, vol. 3, pp. 18–23, Jan. 2015.

[4] L. M. C. Oliveira, R. Dias, C. M. Rebello, M. A. F. Martins, A. E. Rodrigues, A. M. Ribeiro, and I. B. R. Nogueira, "Artificial intelligence and cyber-physical systems: A review and perspectives for the future in the chemical industry," *AI*, vol. 2, no. 3, pp. 429–443, Sep. 2021.

[5] S. S. Jha, "An overview on the explainability of cyber-physical systems," in *Proc. Int. FLAIRS Conf.*, vol. 35, 2022, pp. 1–4.

[6] N. Alimonda, L. Guidotto, L. Malandri, F. Mercorio, M. Mezzanzanica, and G. Tosi, "A survey on XAI for cyber physical systems in medicine," in *Proc. IEEE Int. Conf. Metrol. Extended Reality, Artif. Intell. Neural Eng. (MetroXRAINE)*, Oct. 2022, pp. 265–270.

[7] S. Sivamohan, S. S. Sridhar, and S. Krishnaveni, "TEA-EKHO-IDS: An intrusion detection system for industrial CPS with trustworthy explainable AI and enhanced krill herd optimization," *Peer-to-Peer Netw. Appl.*, vol. 16, no. 4, pp. 1993–2021, Aug. 2023.

[8] L. Almuqren, M. S. Maashi, M. Alamgeer, H. Mohsen, M. A. Hamza, and A. A. Abdelmageed, "Explainable artificial intelligence enabled intrusion detection technique for secure cyber-physical systems," *Appl. Sci.*, vol. 13, no. 5, p. 3081, Feb. 2023.

[9] A. P. Patil, J. Devarakonda, M. Singuru, S. Tilak, and S. Jadon, "XAI for securing cyber physical systems," in *Proc. 3rd Int. Conf. Secure Cyber Comput. Commun. (ICSCCC)*, May 2023, pp. 671–677.

[10] R. Rajkumar, I. Lee, L. Sha, and J. Stankovic, "Cyber-physical systems: The next computing revolution," in *Proc. Design Autom. Conf.*, Jun. 2010, pp. 731–736.

[11] A. Rai, "Explainable AI: From black box to glass box," *J. Acad. Marketing Sci.*, vol. 48, no. 1, pp. 137–141, Jan. 2020.

[12] I. Ahmed, G. Jeon, and F. Piccialli, "From artificial intelligence to explainable artificial intelligence in Industry 4.0: A survey on what, how, and where," *IEEE Trans. Ind. Informat.*, vol. 18, no. 8, pp. 5031–5042, Aug. 2022.

[13] D. L. Marino, C. S. Wickramasinghe, and M. Manic, "An adversarial approach for explainable AI in intrusion detection systems," in *Proc. 44th Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2018, pp. 3237–3243.

[14] A. M. Roth, J. Liang, and D. Manocha, "XAI-N: Sensor-based robot navigation using expert policies and decision trees," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 2053–2060.

[15] S. R. Islam, W. Eberle, S. K. Ghafoor, and M. Ahmed, "Explainable artificial intelligence approaches: A survey," 2021, arXiv:2101.09429.

[16] G. Sofianidis, J. M. Rožanec, D. Mladenić, and D. Kyriazis, "A review of explainable artificial intelligence in manufacturing," in *Trusted Artificial Intelligence in Manufacturing: A Review of the Emerging Wave of Ethical and Human Centric AI Technologies for Smart Production*, J. Soldatos and D. Kyriazis, Eds. Boston, MA, USA: Now, 2021, ch. 5, pp. 93–113.

[17] B. Pradhan, S. Lee, A. Dikshit, and H. Kim, "Spatial flood susceptibility mapping using an explainable artificial intelligence (XAI) model," *Geosci. Frontiers*, vol. 14, no. 6, Nov. 2023, Art. no. 101625.

[18] A. Dikshit and B. Pradhan, "Interpretable and explainable AI (XAI) model for spatial drought prediction," *Sci. Total Environ.*, vol. 801, Dec. 2021, Art. no. 149797.

[19] A. Abdollahi and B. Pradhan, "Explainable artificial intelligence (XAI) for interpreting the contributing factors feed into the wildfire susceptibility prediction model," *Sci. Total Environ.*, vol. 879, Jun. 2023, Art. no. 163004.

[20] A. Cartolano, A. Cuzzocrea, G. Pilato, and G. M. Grasso, "Explainable AI at work! What can it do for smart agriculture?" in *Proc. IEEE 8th Int. Conf. Multimedia Big Data (BigMM)*, Dec. 2022, pp. 87–93.

[21] N. Almakayeel, S. Desai, S. Alghamdi, and M. R. N. M. Qureshi, "Smart agent system for cyber nano-manufacturing in Industry 4.0," *Appl. Sci.*, vol. 12, no. 12, p. 6143, Jun. 2022.

[22] H. Elhoone, T. Zhang, M. Anwar, and S. Desai, "Cyber-based design for additive manufacturing using artificial neural networks for Industry 4.0," *Int. J. Prod. Res.*, vol. 58, no. 9, pp. 2841–2861, May 2020.

[23] M. Ogunsanya and S. Desai, "Physics-based and data-driven modeling for biomanufacturing 4.0," *Manuf. Lett.*, vol. 36, pp. 91–95, Jul. 2023.

[24] E. K. Došilović, M. Breic, and N. Hlupic, "Explainable artificial intelligence: A survey," in *Proc. 41st Int. Conv. Inf. Commun. Technol., Electron. Microelectron. (MIPRO)*, May 2018, pp. 0210–0215.

[25] D. Weyns, J. Andersson, M. Caporuscio, F. Flammini, A. Kerren, and W. Löwe, "A research agenda for smarter cyber-physical systems," *J. Integr. Design Process Sci.*, vol. 25, no. 2, pp. 27–47, Aug. 2021.

[26] M. C. Zizic, M. Mladineo, N. Gjeldum, and L. Celent, "From Industry 4.0 towards Industry 5.0: A review and analysis of paradigm shift for the people, organization and technology," *Energies*, vol. 15, no. 14, p. 5221, Jul. 2022.

[27] K. Amarasinghe, "Explainable neural networks based anomaly detection for cyber-physical systems," Ph.D. dissertation, Dept. Comput. Sci., Virginia Commonwealth Univ., Richmond, VA, USA, 2019.

[28] A. B. Bakker, E. Demerouti, and A. I. Sanz-Vergel, "Burnout and work engagement: The JD-R approach," *Annu. Rev. Organizational Psychol. Organizational Behav.*, vol. 1, no. 1, pp. 389–411, Mar. 2014.